MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 12AKP7

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus, aluminized screen

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Eb3	Ehk	Zd	Rg1	Eb3/Eb2	Alt
Unit:	V	Vdc	v	Vdc	Vdc (see note 2)	Vdc	Vdc	Meg (see note 3)	Meg	Ratio	ft
Maximum:	6.9	0	1,200	3,300	8,800	17,600	±180	1.0	1.5	3.0	10,000
Minimum:	5.7	-300	---	0	2,500	5,000	---	---	---	1.0	---
TEST CONDITIONS:	6.3	Adjust	---	Adjust	7,500	8,500	---	---	---	---	---

GENERAL:

Qualification - Not required

Ⓐ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX
	Quality conformance inspection, part 1						
(A) Appendix D, 30(a), 40, 70	Visual and mechanical inspection criteria		---	---	---	---	---
5201	Voltage breakdown		0.65	II	---	---	---
5206	Gas "cross"	Ib3 = 50 μAdc (see note 4)	0.65	II	---	---	---
5106	Screen and faceplate blemishes		0.65	II	---	---	---
5223	Modulation	Ib3 = 25 μAdc	0.65	II	ΔEc1	---	45
5231	Spot position (electrostatic deflection)		0.65	II	---	---	15
5231	Spot displacement (leakage)		0.65	II	Displ	---	10
5241	Grid cutoff voltage		0.65	II	Ec1	-160	-240
5251	Grid No. 1 leakage current		0.65	II	---	---	---
5251	Anode No. 1 leakage current		0.65	II	---	---	---
5251	Anode No. 2 leakage current		0.65	II	---	---	---
5221	Screen brightness I	Ib3 = 25 μAdc	0.65	II	---	---	---
	Quality conformance inspection, part 2						
(A) Appendix D, 30(b)	Dimensions	See figure 1	6.5	S3	---	---	---
1111	Base pin solder depth		6.5	S3	---	---	---
1301	Heater current		6.5	S3	If	540	660
(A) 5201	Electrode current (anode No. 1)	Ec1 = 0 (see note 3)	6.5	S3	Ib1	-15	+5
(A) 5201	Electrode current (cathode)	Ib3 = 10 μAdc Ib3 = 25 μAdc	6.5 6.5	S3 S3	Ik Ik	--- ---	30 90
5101	Base alinement (electrostatic types)	+1D2, pin No. 1	6.5	S3	---	---	---
(A) 5101	Side terminal alinement (electrostatic types)	+1D2	6.5	S3	---	---	---
5101	Side terminal and base alinement	Pin No. 1	6.5	S3	---	---	---
5101	Neck and base alinement (electrostatic types)		6.5	S3	---	---	---
5101	Angle between traces		6.5	S3	---	89	91

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THOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued							
5216	Stray light emission	Eb2 = 8,800 Vdc; Eb3 = 17,600 Vdc	6.5	S3	---	---	---	---
5221	Screens	Eb2 = 7.5 kV; Eb3 = 8.5 kV; Cb5 = 450 cb (min); G5:1 = 4.0 (min)	6.5	S3	---	---	---	---
5226	Line width "A" (electrostatic deflection)	Ib3 = 25 µAdc	6.5	S3	Width	---	0.43	mm
5226	Line width "B" (electrostatic deflection)	Ib3 = 25 µAdc	6.5	S3	Width	---	0.60	mm
5246	Focusing voltage at cutoff		6.5	S3	Eb1	1,800	2,650	Vdc
5246	Focusing voltage (zero-bias)	See note 3	6.5	S3	Eb1	1,800	2,650	Vdc
5248	Deflection factor (1D2)		6.5	S3	DF	150	230	Vdc/in.
5248	Deflection factor (3D4)		6.5	S3	DF	135	195	Vdc/in.
5251	Heater-cathode leakage current		6.5	S3	---	---	---	---
	Quality conformance inspection, part 3							
---	Life-test provisions	Group C; Eb3 = 17,600 Vdc; Eb2 = 8,800 Vdc; Ib3 = 25 µAdc; t = 500 hours (min)	---	---	---	---	---	---
---	Life-test end points	Ib3 = 25 µAdc	---	---	---	---	---	---
		Line width "A"	---	---	Width	---	0.45	mm
		Line width "B"	---	---	Width	---	0.65	mm
		Modulation	---	---	ΔEc1	---	50	Vdc
	Periodic-check tests							
5101	Neck and bulb alinement (electrostatic types)		---	---	---	---	0.25	inches
5248	Deflection factor uniformity		---	---	---	---	---	---
1141	Pressure (implosion)		---	---	---	---	---	---
5111	Vibration		---	---	Width	---	2.0	mm
1331	Direct-interelectrode capacitance	k to all	---	---	Ck	---	8.0	pF
		g1 to all	---	---	Cg1	---	8.0	pF
		D1 to D2	---	---	C1D2	---	4.0	pF
		D3 to D4	---	---	C3D4	---	3.0	pF
		D1 to all	---	---	CD1	---	14.0	pF
		D2 to all	---	---	CD2	---	14.0	pF
		D3 to all	---	---	CD3	---	10.0	pF
		D4 to all	---	---	CD4	---	10.0	pF

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NOTES:

1. Accelerator power input (average) shall be limited to 6 watts.
2. It is recommended that the deflection electrode circuit resistance be approximately equal. Higher resistance values up to 5.0 megohms may be used for low-beam current operation.
3. Accelerator power input (average) shall be limited to 6 watts to protect the tube against damage. This measurement should be taken by pulsing the grid from cutoff to the specified beam current at a 10 percent duty cycle and multiplying the current meter reading by 10.
4. This test to be performed at the conclusion of the holding period.

The margins of this tube specification sheet are marked with a revision letter (A) to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

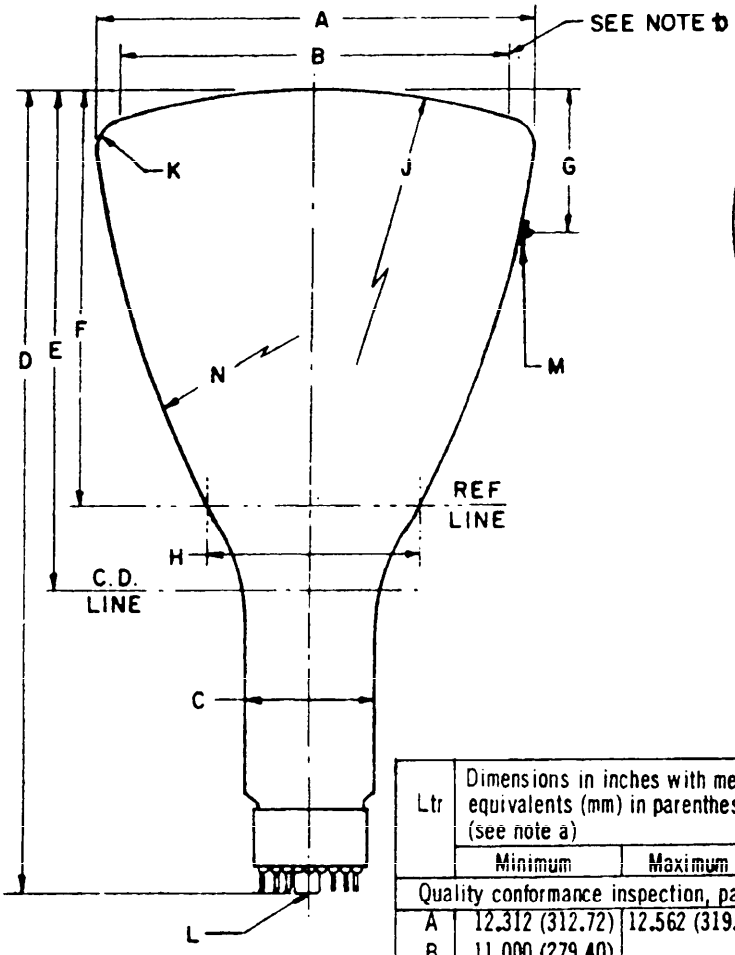
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Air Force - 11

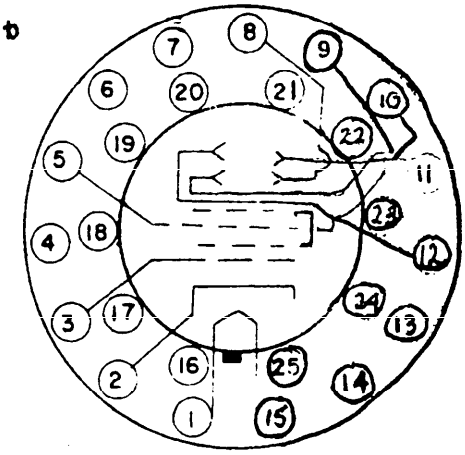
Review activities:
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BOTTOM VIEW OF BASE

PIN NO	ELEMENT
1	HEATER
2	CATHODE
3	GRID NO 1
5	FOCUSING ELECTRODE
8	D4
9	D3
10	ACCELERATOR
11	D2
12	D1
15	HEATER

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note a)	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	12.312 (312.72)	12.562 (319.07)
B	11.000 (279.40)	
C	3.687 (93.65)	3.812 (96.82)
D	23.125 (587.38)	23.875 (606.43)
F	12.094 (307.19)	12.469 (316.71)
G	3.500 (88.90)	4.000 (114.30)
H		5.750 (146.05)
Quality conformance inspection, part 3 (periodic check)		
L	Base: B25-139	
M	Bulb contact: J1-21	
Reference dimensions (see note c)		
E	4.375 (111.13)	
J	40.000 (1016.00) R	
K	.437 (11.10) R	
N	24.000 (609.60) R	

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NOTES:
a. Metric equivalents (to the nearest .01 mm) are given for general information only and are bas[illegible]
b. Useful screen diameter.
c. Reference dimensions are provided for information only; do not use for inspection purpo[illegible]

FIGURE 1. Outline drawing of electron tube type 12AKP7.

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